

$$XGB \text{ (endogenous)} \cdot +E \text{ (earnings)} \cdot +LG \text{ (leaf-graph)} = VolTree$$

OHLCV (ticker i)
 $\rightarrow pk_{i,t}$

earnings calendar
 (external)

endogenous $\mathbb{R}^8$
 (HAR lags + momentum)

earnings $\mathbb{R}^4$
 (distance to release)

$x_{i,t}$

Gamma-loss XGBoost

300 trees · 31 leaves · reg:gamma (QLIKE-equivalent) · 3-seed

per-stock forecast $\sigma^2_{i,t+h}$
 $h \in \{1, 5, 10, 22\}$

$\hat{y}_i$ (all tickers)

leaf-cooccurrence graph smoothing

kNN on shared
 leaves ($k = 10$)

$$\hat{y}_i^{\text{sm}} = (1 - \alpha) \hat{y}_i + \alpha \cdot \text{mean}_{\text{kNN}(i)} \hat{y}$$

α fit on validation